R2160 FLIGHT MANUAL

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Type Certificate nr 70 dated 14.06.1977

Serial Number

Registration EXAMPLE

VISA D G A C

~~Approved sections:~~
~~2, 3, 4 and Section~~ 5
~~pages 5.02 and 5.03~~

This publication includes the material required to be furnished to the pilot by FAR Part 23 and AIR 2052

This aircraft must be operated within the operating limits specified in the present flight manual.

THIS DOCUMENT MUST BE PERMANENTLY KEPT ON BOARD THE AIRCRAFT.

This issue is applicable from S/N 265 included

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niet relevant

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LIST OF REVISIONS

Rev. n°	Pages revised	Description	Date of approval
1	Cover page, ii, iii, 1.06, 1.08, 1.09, 2.01, 2.02, 2.03, 2.04, 2.05, 2.06, 3.02, 3.03, 3.04, 3.05, 3.08, 4.03, 4.05, 4.06, 4.07, 4.09, 4.10, 4.12, 4.13, 4.16, 5.02, 5.05, 5.07, 5.08, 7.01, 7.04, 7.18, 7.19, 7.20, 7.21, 7.22, 7.23, 7.24	- Corrections - 3 position flap control - Use in Utility category - Supplement 3: Aircraft without wheel spats § limitations - Supplement 4: GPS - Supplement 5: Enlarged canopy - Supplement 6: Low voltage	20.01.98

SECTION 0 GENERAL

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LIST OF ABBREVIATIONS USED

sq ft.....Square foot
ft.....Foot
In.....Inches
Nm.....Nautic mile
km.....Kilometer
m.....Meter
cm.....Centimeter
kt.....Knot
m/s.....Meter per second
rpm.....Revolution per minute
Va.....Manoeuvering speed
VC.....Calibrated airspeed
Vfe.....Maximum Flaps Extended speed
Vne.....Never Exceed Speed
Vno.....Maximum cruising speed
Vso.....Stalling speed landing position
Vs1.....Stalling speed flaps up configuration
VI.....Indicated airspeed
km/h.....Kilometer per hour
HP.....Horse Power
hPa.....Hectopascal
in.Hg.....Inches of mercury
mbar.....Millibar
Zp.....Pressure altitude
l.....Liter
Imp gal.....Imperial gallon
us gal.....US gallon
psi.....Pound per square inch
lb.....Pound
kg.....Kilogramme
°C.....Degrees Celcius
°F.....Degrees Farenheit
V.....Volt
A.....Ampere

LIST OF RADIO ABBREVIATIONS

ADF ..... Automatic Direction Finder
ATC..... Air Traffic Control
COM ..... Communication Transceiver
DME..... Distance Measuring Equipment
ELT ..... Emergency Locator Transmitter
IFR ..... Instrument Flight Rules
ILS ..... Instrument Landing System
MKR..... Marker Beacon Receiver
NAV ..... Navigation Indicator and Receiver
AUDIO ..... Audio Control Panel
VFR..... Visual Flight Rules
VHF ..... Very High Frequency
VOR ..... Visual Omni-Range (beacon)

BAROMETRIC PRESSURE CONVERSION TABLE

Below pressure in MILLIBAR or HECTOPASCAL, the pressure in INCHES of MERCURY is indicated.

→mbar or hPa

950	960	970	980	990	1000	1010	1020	1030	1040
28.05	28.35	28.64	28.94	29.23	29.53	29.63	30.12	30.42	30.71
951	961	971	981	991	1001	1011	1021	1031	1041
28.08	28.38	28.67	28.97	29.26	29.56	29.85	30.15	30.45	30.74
952	962	972	982	992	1002	1012	1022	1032	1042
28.11	28.41	28.70	29.00	29.29	29.59	29.88	30.18	30.47	30.77
953	963	973	983	993	1003	1013	1023	1033	1043
28.14	28.44	28.73	29.03	29.32	29.62	29.91	30.21	30.50	30.80
954	964	974	984	994	1004	1014	1024	1034	1044
28.17	28.47	28.76	29.06	29.35	29.65	29.94	30.24	30.53	30.83
955	965	975	985	995	1005	1015	1025	1035	1045
28.20	28.50	28.79	29.09	29.38	29.68	29.97	30.27	30.56	30.86
956	966	976	986	996	1006	1016	1026	1036	1046
28.23	28.53	28.82	29.12	29.41	29.71	30.00	30.30	30.59	30.89
957	967	977	987	997	1007	1017	1027	1037	1047
28.26	28.56	28.85	29.15	29.44	29.74	30.03	30.33	30.62	30.92
958	968	978	988	998	1008	1018	1028	1038	1048
28.29	28.58	28.88	29.18	29.47	29.77	30.06	30.36	30.65	30.95
959	969	979	989	999	1009	1019	1029	1039	1049
28.32	28.61	28.91	29.20	29.50	29.80	30.09	30.39	30.68	30.98

REMINDER:
The Standard pressure of 1013.2 mbar or hPa equals 29.92 in.Hg

FACTORS OF CONVERSION

nautic mile	X	1.852	=	kilometers
feet	X	0.305	=	meters
inches	X	0.0254	=	meters
inches	X	25.4	=	millimeters
feet/minute	X	0.00508	=	meter/second
gallons (US)	X	3.785	=	liters
gallons (Imp)	X	4.546	=	liters
quarts (US)	X	0.946	=	liters
knots	X	1.852	=	km/h
psi	X	0.0689	=	bar
in.Hg	X	33.86	=	mbar
lb	X	0.453	=	kg
(°F - 32)	X	5/9	=	°C

kilometers	X	0.539	=	nautic mile
meters	X	3.281	=	feet
meters	X	39.37	=	inches
millimeters	X	0.03937	=	inches
meter/second	X	197	=	feet/minute
liters	X	0.264	=	gallons (US)
liters	X	0.220	=	gallons (Imp)
liters	X	1.057	=	quarts (US)
km/h	X	0.539	=	knots
bar	X	14.51	=	psi
mbar	X	0.02953	=	in.Hg
kg	X	2.205	=	lb
°C	X	9/5 + 32	=	°F

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SECTION 1 DESCRIPTION

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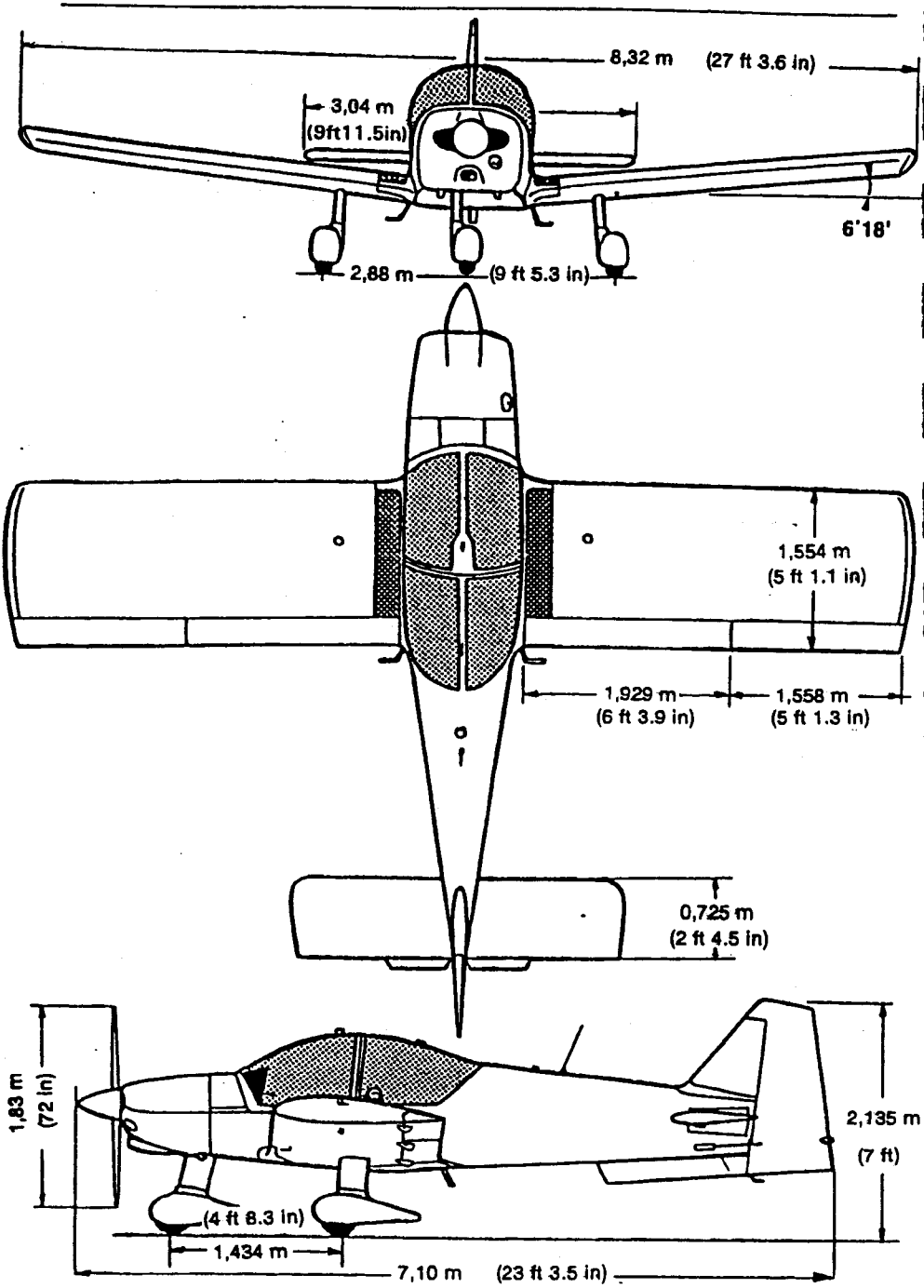

8,32 m
(27 ft 3.6 in)
3,04 m
(9ft11.5in)
6°18'
2,88 m
(9 ft 5.3 in)
1,554 m
(5 ft 1.1 in)
1,929 m
(6 ft 3.9 in)
1,558 m
(5 ft 1.3 in)
0,725 m
(2 ft 4.5 in)
1,83 m
(72 in)
2,135 m
(7 ft)
(4 ft 8.3 in)
1,434 m
7,10 m
(23 ft 3.5 in)

OVERALL DIMENSIONS

Wing span ..... (27 ft 3.6 in) 8,32 m
Overall length ..... (23 ft 3.5 in) 7,10 m
Overall height ..... (7 ft) 2,135 m

INTERNAL CABIN DIMENSIONS

Length ..... (6 ft 8.7 in) 2,05 m
Width ..... (3 ft 5.7 in) 1,06 m
Height ..... (4 ft 1.2 in) 1,25 m
2 seats, accessible from both sides by sliding canopy.

WINGS

Wing area ..... (140 sq ft) 13 m^2
Airfoil ..... NACA 23015
Aspect ratio 5.42
Wing setting ..... 6°18'

AILERONS

Slotted type
Surface (each) ..... (5.54 sq ft) 0,515 m^2

WING FLAPS

Surface (each) ..... (6.8 sq ft) 0,635 m^2
Span (each) ..... (6 ft 3.9 in) 1,929 m

HORIZONTAL STABILIZER

Total control surface ..... (25.2 sq ft) 2,35 m^2
of which anti-balance tab ..... (2 x 0.6 sq ft) 2 x 0,063 m^2
Span ..... (9 ft 11.5 in) 3,04 m

VERTICAL STABILIZER

Surface overall ..... (9.9 sq ft) 0,92 m^2
Stabilizer ..... (3.7 sq ft) 0,35 m^2
Rudder ..... (6.1 sq ft) 0,57 m^2

LANDING GEAR

Fixed Tricycle Type

Track ..... (9 ft 5.3 in) 2,88 m
Wheel base ..... (4 ft 8.3 in) 1,434 m
Tyre size ..... 380 x 150

Oil, shock struts ..... MIL. H. 5606 - A
NORME AIR 3520

Nose gear

Tyre pressure ..... (23 psi) 1,6 bar
Shock strut pressure ..... (58 psi) 4 bar

Main landing gear

Tyre pressure ..... (26 psi) 1,8 bar
Shock strut pressure..... (116 psi) 8 bar

BRAKES

The disk brakes are operated by an independant hydraulic circuit on each main gear wheel.

Hydraulic oil ..... MIL.H.5606 - A
Norme AIR 3520

POWER PLANT

Engine

Manufacturer..... LYCOMING
Type..... O-320-D2A
Number of cylinders 4
Maximum continuous power ..... (160 BHP) at 2700 rpm

Propeller

Manufacturer..... SENSENICH
Type..... 74-DM-6S5-2-64
Diameter ..... 1,83 m (72 in)*
Pitch..... (64 in) 1,62 m
Minimal static rpm full throttle at sea level ..... 2150 rpm
Maximum rpm ..... 2700 rpm
Maximum normal operating rate ..... 2600 rpm

*** Any reduction in diameter during repair is forbidden**

FUEL

Aviation petroleum* ..... AVGAS 100 LL
Fuel grade* ..... (octane) 100 minimum
Total fuel capacity ..... (26.4 imp/31.7 us gal) 120 l
Total usable fuel ..... (26 imp/31 us gal) 118 l
Unusable fuel ..... (0.4 imp/0.7 us gal) 2 l

Single fuselage tank (optional):
Total fuel capacity ..... (35.2 imp/42.3 us gal) 160 l
Total usable fuel ..... (34.7 imp/41.7 us gal) 158 l
Unusable fuel ..... (0.4 imp/0.7 us gal) 2 l

OIL**

Total engine capacity ..... (8 US quarts) 7,5 l
Usable capacity ..... (6 US quarts) 5,7 l

During the first 50 hours of operation:
Pure mineral oil

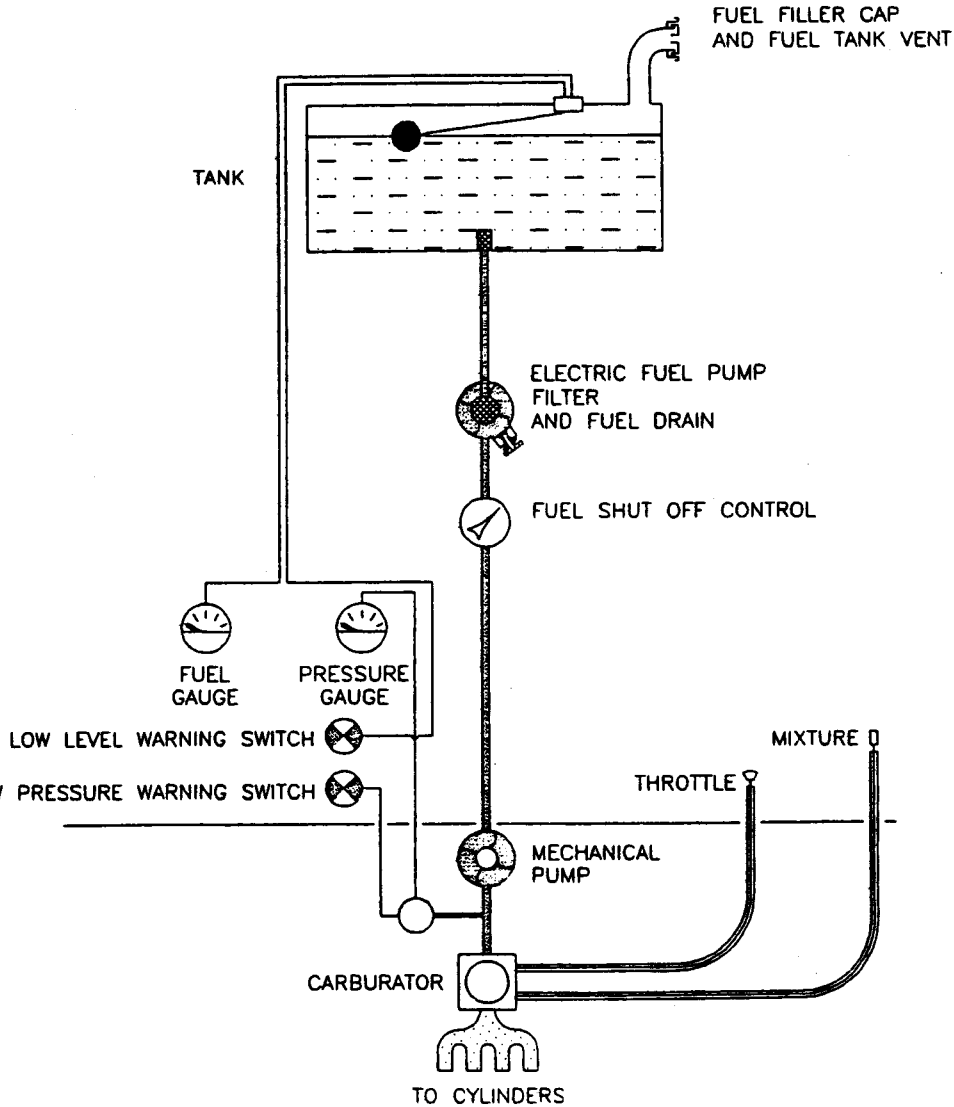
After the first 50 hours of operation:
Dispersant oil

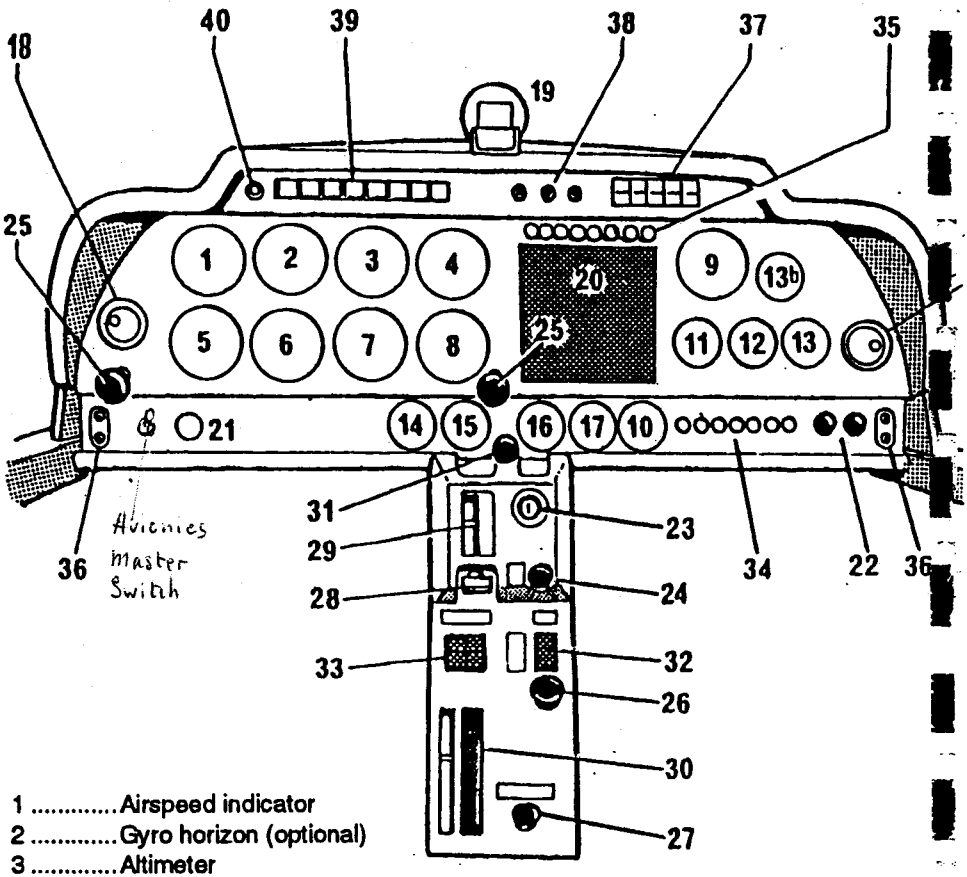
Grades

Oil	Dispersant	Pure mineral
All temperatures	SAE 15W50 or 20W50	----------
Above +25°C (80°F)	SAE 60	SAE 60
Above +15°C (60°F)	SAE 40 or SAE 50	SAE 50
From 0°C to +30°C (30°F to 90°F)	SAE 40	SAE 40
From -15°C to +20°C (0°F to 70°F)	SAE 40, 30 or 20W40	SAE 30
Below -10°C (10°F)	SAE 30 or 20W30	SAE 20

* Refer to Service Instruction Lycoming n°1070 latest edition
** Refer to Service Instruction Lycoming n°1014 latest edition

FUEL SYSTEM


FUEL FILLER CAP
AND FUEL TANK VENT
TANK
ELECTRIC FUEL PUMP
FILTER
AND FUEL DRAIN
FUEL SHUT OFF CONTROL
FUEL
GAUGE
PRESSURE
GAUGE
LOW LEVEL WARNING SWITCH
OW PRESSURE WARNING SWITCH
MIXTURE
THROTTLE
MECHANICAL
PUMP
CARBURATOR
TO CYLINDERS



1 .....Airspeed indicator
2 .....Gyro horizon (optional)
3 .....Altimeter
4 .....~~Optional equipment~~ VOR Indicator
5 .....Turn coordinator (turn and bank indicator optional)
6 .....Directional gyro (optional)
7 .....Rate of climb indicator (optional)
8 .....Tachometer (or optional equipment)
9 .....Optional equipment (or tachometer)
10 .....Fuel pressure
11 to 13 ..Optional equipment
14 .....Oil pressure
15 .....Oil temperature
16 .....Fuel quantity indicator
17 .....Voltmeter
18 .....Cabin ventilators
19 .....Magnetic compass
20 .....COM/NAV ~~(optional)~~ GPS

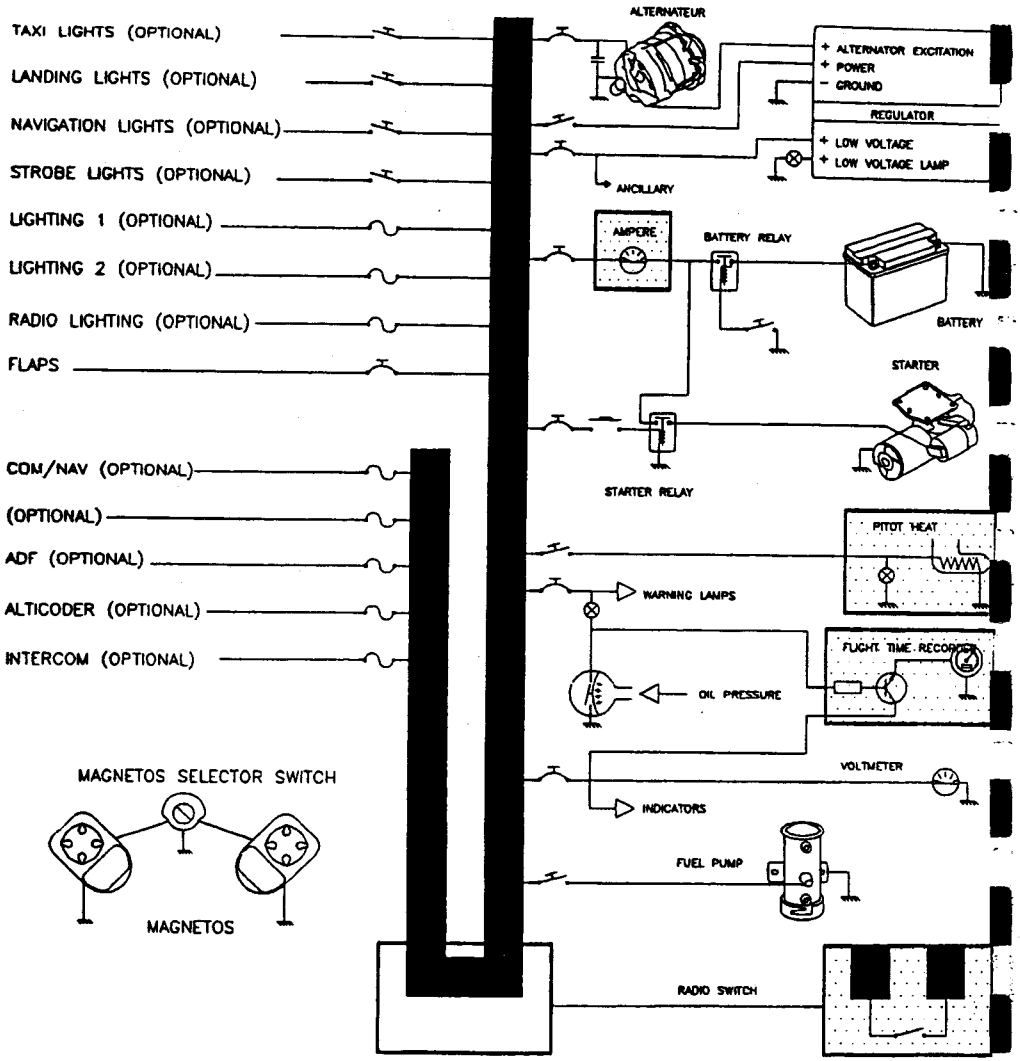
11. G-meter
12. OAT (outside air temperature)
13 GPS data loader plug
13a Flight hours recorder

21 .....Vacuum gauge (optional)
22 .....Cabin heat control
23 .....Magnetos selector switch with starter
24 .....Mixture control
25 .....Throttle control
26 .....Carburetor heat control
27 .....Fuel shut off control
28 .....Flaps control lever
29 .....Flaps position indicator
30 .....Elevator trim tab control wheel
31 .....Parking brake control
32 .....Electric fuel pump switch
33 .....Battery and alternator switch
34 .....Circuit breakers (from L to R):
- Ancillary systems
- Flaps
- Starter
- Indicators
- Warning lights
- Battery
- Charge

35 .....Fuses (according to optional equipment installed):
COM/NAV, ATC, alticoder, Intercom, option, panel lighting (under glareshield), overhead flood-lights, radio bay lighting
36 .....Mike, Headset-Jack sockets
37 .....Safety switch (from L to R):
landing -, taxi-, strobe and navigation lights (option), pitot heat (option)
38 .....Panel lighting controls (from L to R):
1/ on glareshield 2/ panel overhead flood light 3/ radio bay equipment
39 .....Warning lamps (from L to R):
low oil pressure, low fuel pressure, low fuel level, charge, starter engaged, flaps down, 2 warning lamps unused
40 .....Warning lamps test and dimming control

Nota:
Accelerometer: 11 to 13 (or other)

ELECTRICAL SYSTEM



HEATING AND VENTILATION

A - Collector/heat exchanger

B - Cabin heat/off control box

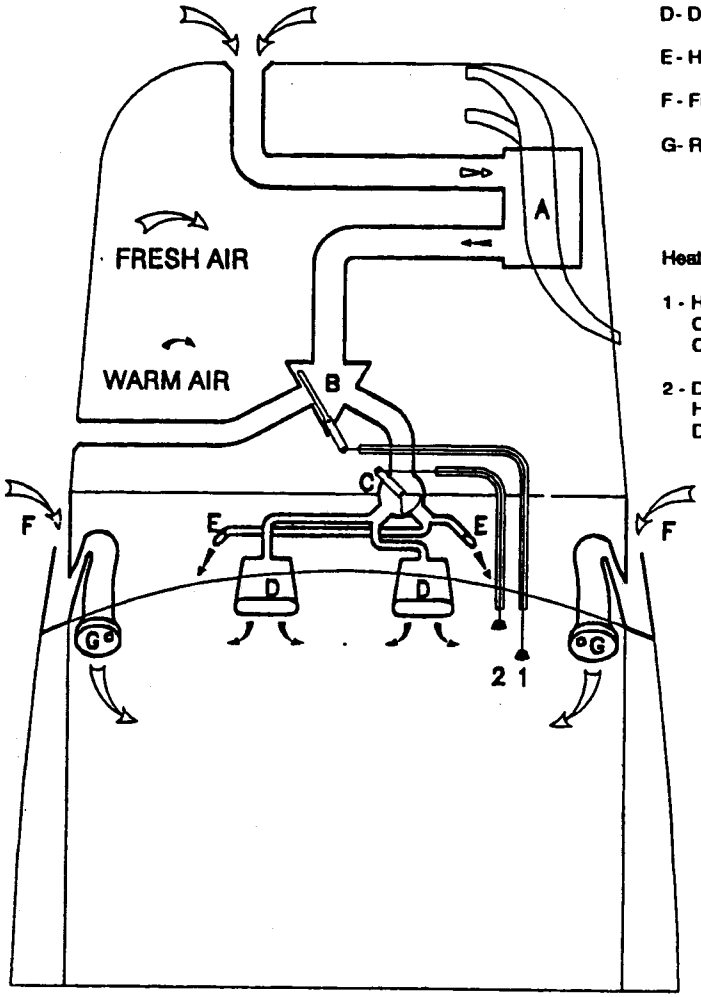
C - Defrost/heating selection box

D - Defrost vents

E - Heating vents

F - Fresh air intake

G - Rotating cabin vents



Heating controls

1 - Heating control:

On	Pull out
Off	Push in

2 - Defrost/heating selection:

Heating	Pull out
Defrost	Push in

CERTIFICATION STANDARDS

The R2160 aircraft has been certified on 14.06.1977 in the "Acrobatic" and "Utility" categories conforming to following technical conditions:

- Standard technical conditions: FAR 23, Amendment 1 to 9 included
- Complementary conditions AIR 2052, § 3.397 and 3.399
- Special condition: the canopy must be jettisonable.

NOTE

All speeds in this manual are Indicated Airspeed unless otherwise specified.

APPROVED OPERATION

VFR by day in non-icing conditions

AIRSPEED LIMITATIONS	km/h	kt
Vne *(never exceed)*	331	178.5
Vno *(max. cruise)*	236	127
Va *(max. maneuver)*	236	127
Vfe *(max. flaps extended)*	180	97

AIRSPEED INDICATOR MARKINGS		km/h	kt
Red line *(never exceed)*	Vne	331	178.5
Yellow arc *(operate with caution and only in "smooth air")*	Vno - Vne	236 - 331	127 - 178.5
Green arc *(normal operating range)*	Vs1 - Vno	117 - 236	63 - 127
White arc	Vso - Vfe	94 - 180	51 - 97

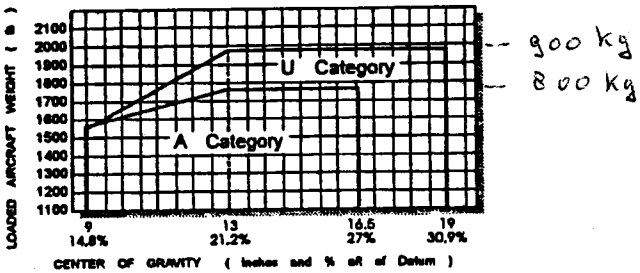
LOAD FACTOR LIMITS AT GROSS WEIGHT

	U category	A Category
Flaps up	n = + 4,4	n = + 6
	n = - 1,76	n = - 3
Flaps down	n =+ 2	n = + 2

MAXIMUM AUTHORIZED WEIGHTS IN "U" CAT.

	U Category	A Category
On take off	(1984 lb) 900 kg	(1764 lb) 800 kg
On landing	(1984 lb) 900 kg	(1764 lb) 800 kg

WEIGHT AND BALANCE



Levelling ..... upper fuselage longeron
Datum ..... leading edge at chord n°5
Chord line ..... (61.2 in) 1.554 m

LOAD PLANNING (Refer also to weight and balance chart, section 6)

The weight of engine oil, as well as unusable fuel, must be included in the empty weight of the aircraft.

	Weight kg (lb)	Arm m (in)
Pilot and passenger	2 x 86 (2 x 190) **	0,46 (18.1)
Fuel	85 (187)	1,115 (43.8)
Baggage *	35 (77)	1,21 (47.6)

* Within the authorized weight and balance limits
** In A category

ENGINE LIMITATIONS

Continuous starter operation ..... 15 to 20 sec.
Maximum power (red line) ..... 2700 rpm

TACHOMETER MARKINGS

Green arc..... 2300 to 2600 rpm
Red line ..... 2700 rpm

FUEL

Aviation petroleum * ..... AVGAS 100 LL
Fuel grade * ..... (octane) 100 minimum
Total fuel capacity..... (26.4 imp/31.7 us gal) 120 l
Total usable fuel ..... (26 imp/31 us gal) 118 l
Unusable fuel ..... (0.4 imp/0.7 us gal) 2 l
Normal pressure ..... (0.5 to 5 psi) 35 to 350 mbar

~~Single fuselage tank (optional):~~
~~Total fuel capacity..... (35.2 imp/42.24 us gal) 160 l~~
~~Total usable fuel ..... (34.7 imp/41.7 us gal) 158 l~~
~~Unusable fuel ..... (0.4 imp/0.7 us gal) 2 l~~

OIL

Maximum temperature (red line) ..... (245°F) 118°C
Normal temperature (green arc) ..... (140 to 245°F) 60 to 118°C
Minimal idle pressure (red line) ..... (25 psi) 1,70 bar
Yellow arc ..... (25 to 55 psi) 1,70 to 3,8 bar
Normal pressure (green arc) ..... (55 to 95 psi) 3,8 to 6,6 bar
Yellow arc (ground warm up)..... (95 to 115 psi) 6,6 to 7,9 bar
Maximum pressure cold start and take off (red line) ... (115 psi) 7,9 bar

* Refer to Service Instruction Lycoming n° 1070 latest edition.

LOAD LIMITS

Number of occupants 2
Maximum authorized weight of baggage (in U cat.)..... (77 lb) 35 kg

OPERATIONAL LIMITS IN "U" CATEGORY

Within the limits of this category, the following manoeuvres are authorized:

Turn at more than 60° bank..... entry speed (108 kt) 200 km/h
Lazy eight ..... entry speed (130 kt) 240 km/h
Chandelle ..... entry speed (130 kt) 240 km/h

ENGINE FAILURE DURING TAKE OFF (roll)

With sufficient runway remaining

Throttle ..... idle (pull)
Brakes ..... as required
Mixture ..... off (pull out)
Magneto switch ..... off
Alternator switch ..... off
Battery switch ..... off

Without sufficient runway remaining

Throttle ..... idle (pull)
Brakes ..... apply heavily
Mixture ..... off (pull out)
Fuel shut off control ..... pull out
Magneto switch ..... off
Battery switch ..... off

ENGINE FAILURE IMMEDIATELY AFTER TAKE OFF

Glide speed ..... (78 kt) 145 km/h
Mixture ..... off (pull out)
Fuel shut off control ..... pull out
Magneto switch ..... off
Battery switch ..... on (in order to use the flaps)

NOTE CAREFULLY

Land straight ahead, with only small direction changes to avoid obstructions.

Never try to turn back to the runway, as altitude after take off is seldom sufficient.

ENGINE FAILURE IN FLIGHT

If altitude is evaluated to be sufficient to try an engine restart:

Establish maximum glide speed, flaps up 145 km/h (78 kt). In these conditions and without wind, the aircraft covers approximately 8,7 times its altitude.

Fuel shut off control ..... push in
Electric pump ..... on
Mixture ..... full push in
Throttle ..... 1/4 travel forward
Magneto switch ..... sur L+R ("Both")

If the propeller still turns, the engine should restart.

If the propeller is stopped, operate the starter.

If the engine still does not start, prepare for a forced landing, following the procedure below:

POWER OFF FORCED LANDING OFF AIRFIELD

Look for a suitable landing area:

Airspeed ..... (78 kt) 145 km/h
Belts and harnesses ..... tight
Electric pump ..... off
Mixture ..... off (pull out)
Throttle ..... to idle (pull)
Magneto switch ..... off
Fuel shut off control ..... pull out
Alternator switch ..... off

Final

Flaps ..... full down
Battery switch ..... off
Canopy ..... unlocked

PRECAUTIONARY POWER LANDING OFF AIRFIELD

Fly over the choosen field several times at low speed (145 km/h - 78 kt) in order to locate the most suitable landing area, flaps in take off position (10°), then make a precautionary approach at 122 km/h (66 kt), flaps in landing position (35°).

On final, unlock the canopy.

Before touch down

Magneto switch ..... off
Battery switch ..... off

NOTE: IN CASE OF CANOPY JAMMING

Canopy handle in "open" position.

Free the two canopy release levers located on the arm rests, on both sides of the instrument panel and place them in vertical position.

FIRE

Engine fire during starting

Keep the engine turning with:
Fuel shut off control ..... pull out
Electric pump ..... off
Throttle ..... full power (push)
Mixture ..... off (pull out)

The aim of this procedure is to make the engine "swallow" the accumulated fuel in the inlet pipes (generally following an excess of fuel injection during a difficult engine start).

If fire continues

Magneto switch ..... off
Battery switch ..... off
Alternator switch ..... off

Abandon the aircraft and try to extinguish the fire with the aids available: fire extinguishers, covers, clothing or sand.

Engine fire in flight

Fuel shut off control ..... pull out
Fuel throttle ..... full power until engine stops
Mixture ..... off (pull out)
Electric pump ..... off
Alternator switch ..... off
Cabin heat and ventilation ..... off
Speed ..... (86 kt) 160 km/h

Prepare for a forced landing off airfield, following the procedures in the chapter "Power off forced landing off airfield".

Do not attempt to restart the engine.

Cabin fire

Extinguish the fire by all means possible (optional extinguisher).

To eliminate smoke, apply maximum ventilation.

In case of an electrical fire (fumes indicating insulation burning):

Cabin ventilation ..... reduce
Alternator switch ..... off
Battery switch ..... off
Battery circuit breaker ..... pull out
Alternator circuit breaker ..... pull out

Land quickly if the fire continues.

VIBRATIONS AND ENGINE ROUGHNESS OPERATION

Vibrations and engine roughness operation are generally due to (verify in this order):

- Carburetor icing: see paragraph "ICING" on next page

- Mixture set too rich or too lean: adjust the mixture (see section 4)

- Contamination in the fuel system: verify fuel pressure. Switch on the electric fuel pump

- Ignition failure: magneto switch on "L" then on "R", then return to "Both". Select the position providing best engine operation and fly to the nearest airfield, at reduced power and mixture set ~~"full rich"~~. to obtain the smoothest engine operation possible.

LOW OIL PRESSURE

In case of low oil pressure indication, check oil temperature and if is too high (red arc):

- Reduce power

- Fly to the nearest airfield and prepare for an off airfield landing

ICING

Proceed as follows when inadvertently encountering icing:

- Carburetor heat on (pull)
- Increase power in order to reduce ice build-up to minimum
- Switch on pitot heat (if installed)
- Select maximum cabin heat and direct the total output to the windscreen ("defrost" position) in order to remove the ice quickly
- Turn back or change altitude, to obtain an outside air temperature less conducive to icing
- Plan to land at the nearest airfield

With an extremely rapid ice build-up, carry out a forced landing.

Remember that a layer of 0,5 cm (0.2 in) on the wing leading edge fairly increases stall speed. If needed, use a higher than normal approach speed: 140 to 150 km/h (75 to 81 kt).

REMARKS

If continuous carburetor heat is deamed necessary, it is imperative to adjust the mixture control to obtain normal engine operation.

Always use carburetor heat fully on or fully off, in certain cases, an intermediate position could increase icing.

ELECTRICAL POWER SUPPLY MALFUNCTION

Alternator failure is indicated when the amber "alternator failure" light on the warning panel is lit, and a progressive drop in voltage (shown on the voltmeter).

If amber light is lit

Switch off alternator, then back on.
This operation resets the overvoltage relay which may have cut-out due to a transiant overvoltage.

If the failure continues

- Switch off the alternator
- Switch off all electrical equipment not essential for continuing the flight
- Land as soon as possible and have the electrical system inspected

NOTE
An alternator failure does not prevent the engine from operating normally

INADVERTENT SPIN

Should a spin occure, use the following procedure:

- Throttle ..... idle (pull)
- Rudder..... maximum opposite to direction of rotation
- Elevator ..... fully back
- Ailerons ..... neutral
- Once rotation stops, rudder to neutral position and recover within flight limitations.

NOTE
If flaps are down when spin begins, retract them immediately

LOSS OF ELEVATOR CONTROL

In the event of loss of elevator control (accidental disconnection):

- Stabilize the aircraft in level flight, flaps at 35°, at 138 km/h (75 kt), using the elevator trim and throttle

- Do not change the elevator trim setting and control the angle of descent only with throttle. Reduce power only on short final and near to the ground.

SECTION 4 NORMAL PROCEDURES

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LOADING

Before each flight, insure that the total weight and the load balance are within the established limits. For this, use the weight and balance chart in Section 6.

NORMAL OPERATING SPEEDS

The speeds reminded hereunder are Indicated Airspeeds recommended for normal operations.

They are based on standard aircraft, operated at gross weight, in standard atmosphere, at sea level. They can change from one aircraft to another, depending on installed equipment, aircraft and engine condition, atmospheric conditions and pilot proficiency.

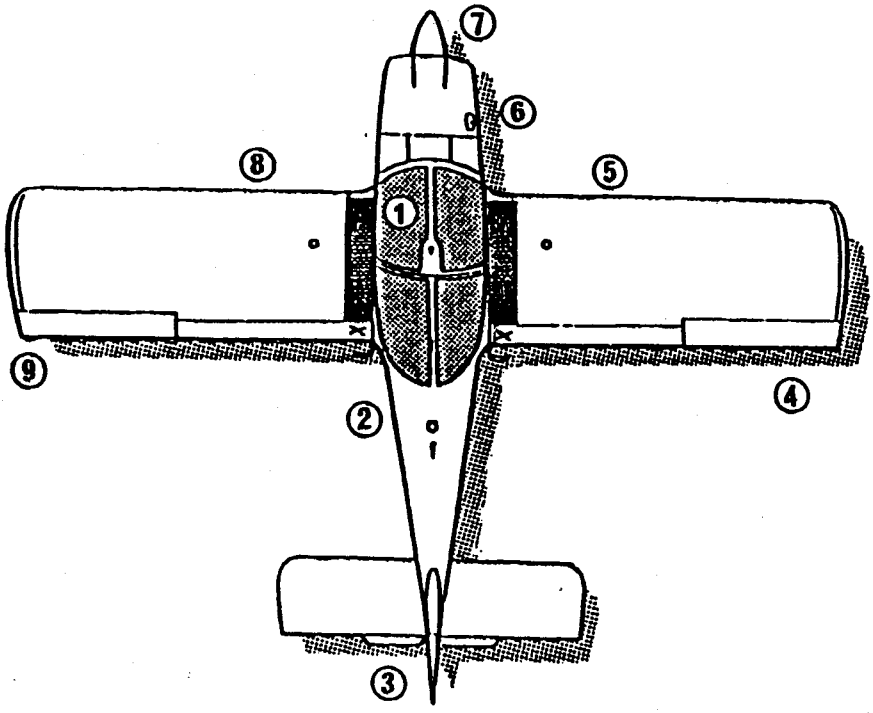
- Best rate of climb speed
  flaps in take off position (10°) ..... (70 kt) 130 km/h
  flaps up ..... (78 kt) 145 km/h

- Best angle of climb speed
  flaps in take off position (10°) ..... (65 kt) 120 km/h
  flaps up ..... (70 kt) 130 km/h

- Maximum operating speed in turbulence
  flaps up ..... (127 kt) 236 km/h

- Maximum speed
  flaps in landing position (35°) ..... (97 kt) 180 km/h

- Landing speed (final approach)
  flaps in landing position (35°) ..... (65 kt) 120 km/h



PRE-FLIGHT INSPECTION

To be performed before each flight.
This inspection may be reduced after intermediate on route landings.

Magneto switch ..... "OFF"
Controls ..... free
Battery switch ..... on
Flaps ..... check operation
Fuel quantity ..... check
Battery switch ..... off
Aircraft documents ..... check availability on board
Baggage ..... check stowing

Check flight controls displacements, then make an aircraft walk-around inspection (as shown above) beginning at the fuselage left side.

	Fuel filler cap	in place, locked
1	Static vent	clean, non obstructed
	Tank drain valve	actuated
2	Horizontal stabilizer	surface condition, hinges without clearance
	Rudder	check hinges and clearance
3	Static vent	clean, unobstructed
4	Flap, aileron	check condition and hinges
	Wing tip, navigations lights if installed (optional)	check condition
	Stall warning	clean, check displacement
5	Right main landing gear	check attachment and fairing condition
		normal shock absorber compression
		tyre inflated
	Oil level	check, oil cap secured, panel closed
	Engine cowl attachment	check
6	Propeller	clean, in good condition
	Propeller spinner	no play
	Air inlets	clean, non obstructed
	Landing light if installed (optional)	window clean
	Nose gear	check attachment and fairing condition
		normal shock absorber compression
7		tyre inflated
		tow bar removed
	Exhaust pipes	rigid
	Canopy cleanliness	check
	Left main landing gear	check attachment and fairing condition
		normal shock absorber compression
8		tyre inflated
	Pitot	clean, non obstructed
9	Wing tip, navigation-, taxi-, landing lights if installed (optional)	check condition
	Flap, aileron	check condition and hinges

TAXIING

Parking brake ..... unlocked
Brakes ..... test
Turn and bank indicator (optional) ..... check
Directional gyro if installed (optional) ..... check setting
Avoid exceeding 1200 rpm as long as oil temperature remains in the yellow range.

ENGINE RUN UP

Parking brake ..... locked
Oil pressure and temperature ..... green range
Fuel pressure ..... green range
Mixture ..... full push in
Carburetor heat ..... full out

Magneto check

Throttle ..... 1800 rpm
Magneto selection:
Max. drop between (L) or (R) and (L+R) ..... 175 rpm
Max. difference between (L) and (R) ..... 50 rpm

Carburetor heat check

Carburetor heat ..... full on
Check rpm drop ..... between 20 and 200 rpm
Carburetor heat ..... full out

Mixture check

Lean until rpm reduction, then return to "full rich".

Engine idle check

Throttle ..... 600 to 650 rpm

BEFORE TAKE OFF

Controls ..... free
Magneto switch ..... L + R ("Both")
Cabin (seats, belts, canopy) ..... check
Fuel shut off control: pushed in and safety wire-locked..... check
Electric pump ..... on
Elevator trim ..... take off position
Instruments ..... check, set
Flaps ..... (10°) take off position
Throttle ..... "holding" 1200 rpm

TAKE OFF

Normal take off

Take off minimum rpm 2150
Rotation speed ..... (49 kt) 90 km/h
Initial climb speed..... (70 kt) 130 km/h

After obstacle clearance,

Reduce angle of climb to obtain ..... (78 kt) 145 km/h
Electric pump ..... off
Fuel pressure ..... check (green range)
Flaps ..... up

Short take off

Flaps ..... (10°) take off position
Apply full power, brakes applied
then release the brakes ..... mini 2150 rpm
Rotation speed ..... (49 kt) 90 km/h
Then, if necessary (to clear an obstacle)
best angle of climb speed..... (65 kt) 120 km/h

Cross wind take off

Flaps ..... take off position (10°)
Ailerons ..... into the wind
Take off at a slightly higher airspeed than normal
Correct drift in the normal way (max. bank angle, close to the ground: 15°)

Demonstrated crosswind velocity: (18 kt) 33 km/h

CLIMB

Normal climb (flaps up)

Set climb speed 145 km/h (78 kt); 130 km/h (70 kt) at 10 000 ft.

Above 5000 ft, adjust mixture.

Best angle of climb

A better rate of climb is obtained at 130 km/h (70 kt) flaps up and 120 km/h (65 kt) flaps in take off position (10°).

NOTE
This type of climb should only be used exceptionally, due to poor engine cooling.

CRUISE

Refer to Section 5 for rpm settings and cruise performance.

Operation of mixture control

Maintain mixture control in the "full rich" position during take off and in the climb.

In certain conditions (high altitude take off, long climb above 5000 ft), this setting could be too rich, and result in irregular engine operation or loss of power.

In these cases, adjust the mixture to recover regular engine operation, and not for fuel economy.

Mixture adjustment when in stable cruise:

Progressively lean the mixture until a slight reduction in rmp is noted; then lightly enrich to re-establish power and normal engine operation.

NOTE

Take care not to lean the mixture too much, which would cause engine overheating.

ALWAYS ENRICH THE MIXTURE BEFORE AN INCREASE IN POWER.

DESCENT

Rapid descent

Power..... as required to maintain the desired descent path
Carburetor heat..... as required hot (pull) or cold (push)
Each 1500 ft, apply power to avoid excessive engine cooling and to clean the spark plugs.

Approach or down wind

Mixture..... fully rich (push in)
Electric pump..... on
Carburetor heat..... as required hot (pull) or cold (push)
Cabin (belts and seats)..... check
Flaps ..... below 180 km/h (97 kt) in take off position (10°)
Speed..... (70 to 76 kt) 130 to 140 km/h
Elevator trim ..... set
Roll stabilizer or autopilot (if installed) ..... off

Final

Carburetor heat..... cold (push)
Flaps ..... landing position (35°)
Approach speed ..... (65 to 68 kt) 120 to 125 km/h
Elevator trim ..... set

LANDING

Short landing

Flaps ..... landing position (35°)
Approach speed ..... (with throttle setting)(62 kt) 115 km/h
After touch down, brake heavily keeping nose up with elevator and retracting flaps.

Landing in crosswind or gusty conditions

Flaps ..... 10°
Approach speed ..... (73 kt) 135 km/h + 1/2 gust speed
Drift ..... correct in the normal way
Demonstrated crosswind ..... (18 kt) 33 km/h

Overshoot procedure

Carburetor heat cold (push in) ..... check
Throttle ..... full (push in)
Speed ..... (65 kt) 120 km/h
Flaps ..... (10°) take off position
Normal climb speed ..... (70 kt) 130 km/h

AFTER LANDING

Electric pump ..... off
Flaps ..... up
Navigation instruments ..... off

ENGINE SHUT-DOWN

Parking brake ..... tight
COM/NAV and electrical equipment ..... off
Canopy ..... closed, locked
Magneto cut-off check at idle ..... off, then L + R ("Both")
RPM 1000
Mixture ..... off (pull out)

After the engine stops

Magneto switch ..... off
Alternator switch ..... off
Battery switch ..... off
When wheel chocks in place ..... release the parking brake

PARKING BRAKE USE

Brake on

Press on both pedals. Keep pressure on, while pulling the parking brake control upwards. Then, release the pressure on the pedals (the parking brake control remains in the pulled position).

Brake off

Push the control down.

AEROBATIC FLIGHTS

IMPORTANT NOTES

- This aircraft is not provided with a fuel system allowing inverted flight.
- The lubrication does not take place while the aircraft is in the inverted position. A special device (optional) is provided to prevent the oil flowing through the engine breather.
- If the aircraft is equipped with a liquid electrolyte type battery, the latter is protected by an appropriate receptable but is not, however, sealed.
- The luggage hold must be empty and no object must be left loose inside the cabin.
- Make sure that the aircraft C of G is within the permissible limits.

Spins

It is recommended to perform the spin in the following manner:

- Flaps must be retracted
- Start the spin at an adequate height above the safety altitude, taking into consideration that the loss of altitude is about 230 ft per revolution and that the final recovery takes about 1300 ft
- Throttle back in level flight, decrease the speed with a slightly positive vertical position
- When closed to the stalling point:
  pull the elevator control fully
  ailerons in neutral and simultaneously
  rudder in the direction of rotation
- When 2 or 3 rotations have been completed, apply the following recovery procedure:
  rudder in fully opposed direction
  elevator control pulled fully
  ailerons in neutral
- When spin rotation stops, and normal angle of attack is resumed, centralise elevators.

Example to recovery from a LH spin

- Apply and maintain full Right rudder, ailerons in neutral
- Stick back
- After 3 revolutions, recovery is performed in approximately three quarter of a revolution.

Only one action is important: keep the rudder fully in opposite direction !

- In a spin of more than 3 revolutions, the engine is likely to stall. This raises no difficulty: the only additional action required is to use the starter during the recovery.
- With a 4 revolution (or more) spin, recovery is performed in 1,5 revolution
- During the recovery phase, keep a watch on the A.S.I and on the accelerometer, to keep the operating limits laid down.

Authorized Aerobatic figures	Initial speed
Positive spin	stall
Positive loop	(129 kt) 240 km/h
Roll	(108 kt) 200 km/h
Stall turn	(119 kt) 220 km/h
45° half roll and dive out	(119 kt) 220 km/h
Half loop and roll out	(135 kt) 250 km/h
Flick roll	(81 to 86 kt) 150 to 160 km/h max (92 kt) 170 km/h
Chandelle	(119 kt) 220 km/h
Lazy eight	(119 kt) 220 km/h
Turns at more than 60° bank	(108 kt) 200 km/h

If, during one of the figures the engine stalls, it is preferable to throttle during the recovery only. The above figures can be performed without unpriming the engine and at a load factor not exceeding 4 g.

Inverted flight

For aircraft equipped with a dry battery and an inverted fuel system supply the inverted flight is allowed during 20 seconds.

In order to prevent engine roughness apply following procedure:

Entry speed ..... (119 kt) 220 km/h
As aircraft becomes inverted ..... close throttle
For 20 seconds only ..... maintain speed above (80 kt) 150 km/h
After return to normal flight ..... open throttle smoothly

SECTION 5 PERFORMANCE

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NOISE LIMITATION

In compliance with the decree of 3.04.1980, the maximum acceptable noise level for the R2160 aircraft, at a certified gross weight of (1984 lb) 900 kg, is 72 dB(A) (ICAO annex 16, chapter 6).

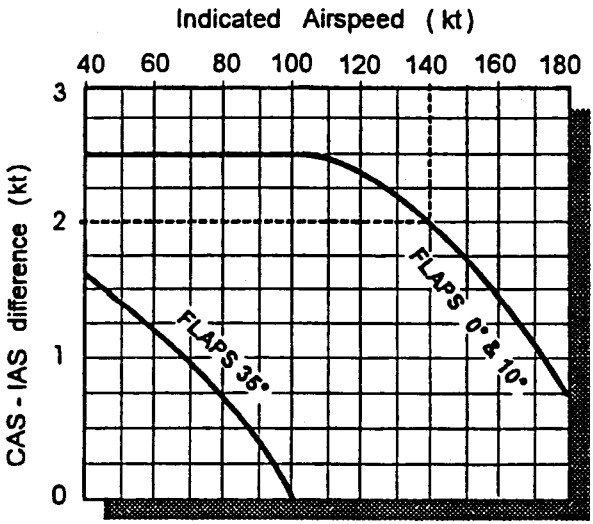
The noise level determined under the conditions of the decree is 69,8 dB(A).

The R2160 aircraft has received noise limitation certificate Nr N70.

STALL SPEEDS

Engine idling, weight: 900 kg (1984 lb)	km/h (kt)		
Bank angle	0°	30°	60°
Flaps up	117 (63)	126 (68)	165 (89)
Flaps 10°, take off position	113 (61)	121 (66)	160 (86)
Flaps 35°, landing position	94 (51)	101 (55)	133 (72)

AIRSPEED INSTALLATION CALIBRATION



NOTE

All speeds in this manual are Indicated Airspeeds unless otherwise specified

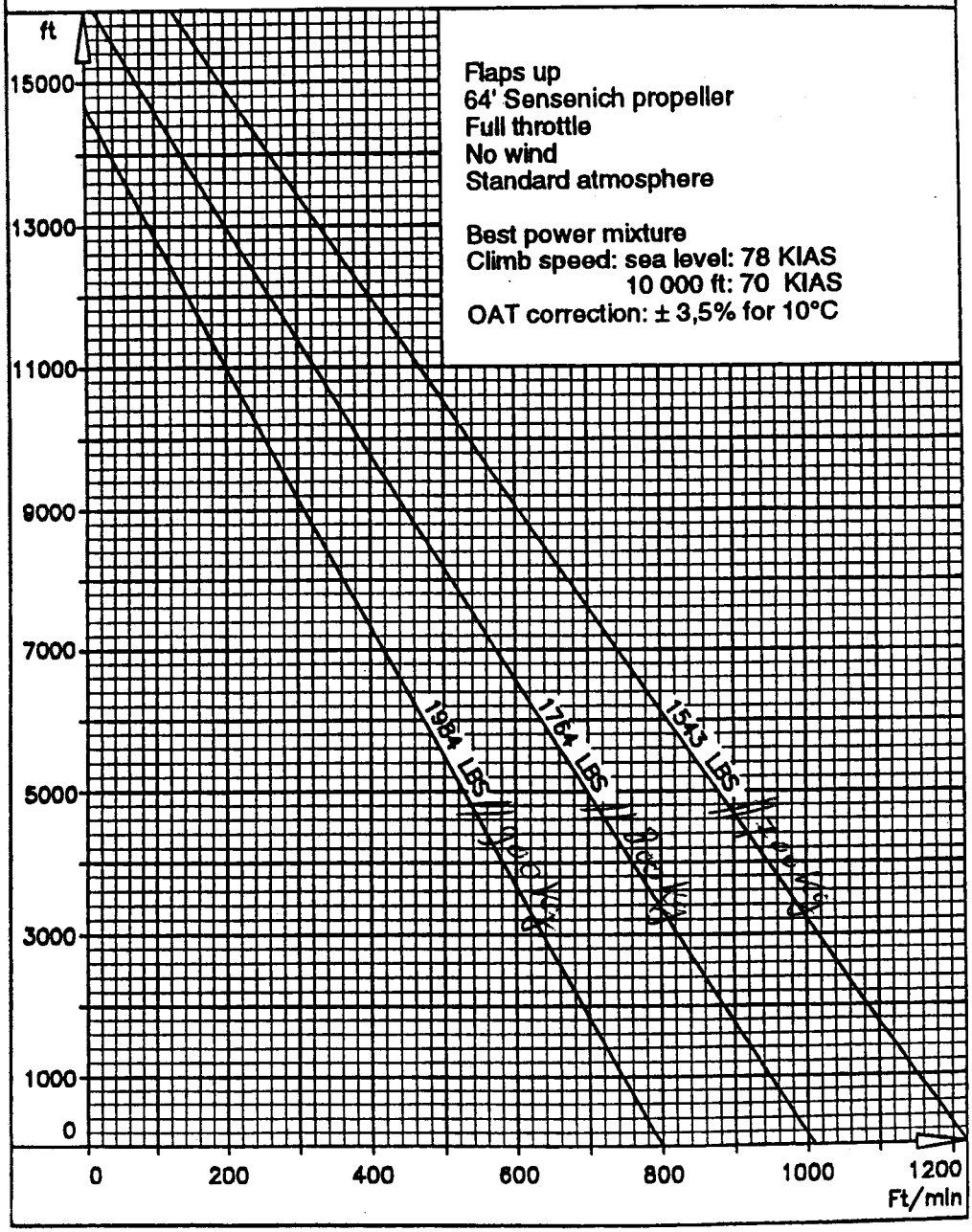
TAKE OFF PERFORMANCE

Concrete level dry runway. Flaps in take-off position. Full throttle							
Max. weight kg (lb)	Head wind (kt)	Sea level + 15°C		2500 ft - 760 m + 10°C		5000 ft - 1525 m + 5°C	
		Run m (ft)	15 m (50 ft) clear	Run m (ft)	15 m (50 ft) clear	Run m (ft)	15 m (50 ft) clear
900 (1984)	0	320 (1050)	574 (1883)	400 (1312)	700 (2297)	490 (1608)	870 (2854)
	10	224 (735)	476 (1562)	280 (919)	590 (1936)	340 (1116)	730 (2395)
	20	147 (482)	385 (1263)	180 (590)	470 (1542)	224 (735)	590 (1936)
800 (1764)	0	230 (755)	410 (1345)	285 (935)	500 (1640)	350 (1148)	620 (2034)
	10	160 (525)	340 (1115)	200 (656)	420 (1378)	245 (804)	520 (1706)
	20	105 (344)	275 (902)	130 (427)	335 (1099)	160 (525)	420 (1378)
700 (1543)	0	160 (525)	285 (935)	200 (656)	350 (1148)	245 (804)	430 (1411)
	10	110 (361)	240 (787)	140 (459)	290 (951)	170 (558)	360 (1181)
	20	70 (230)	190 (623)	90 (295)	235 (771)	110 (361)	290 (951)

Increase distances by 8% for every 10°C increase of the standard temperature, at the altitude concerned.

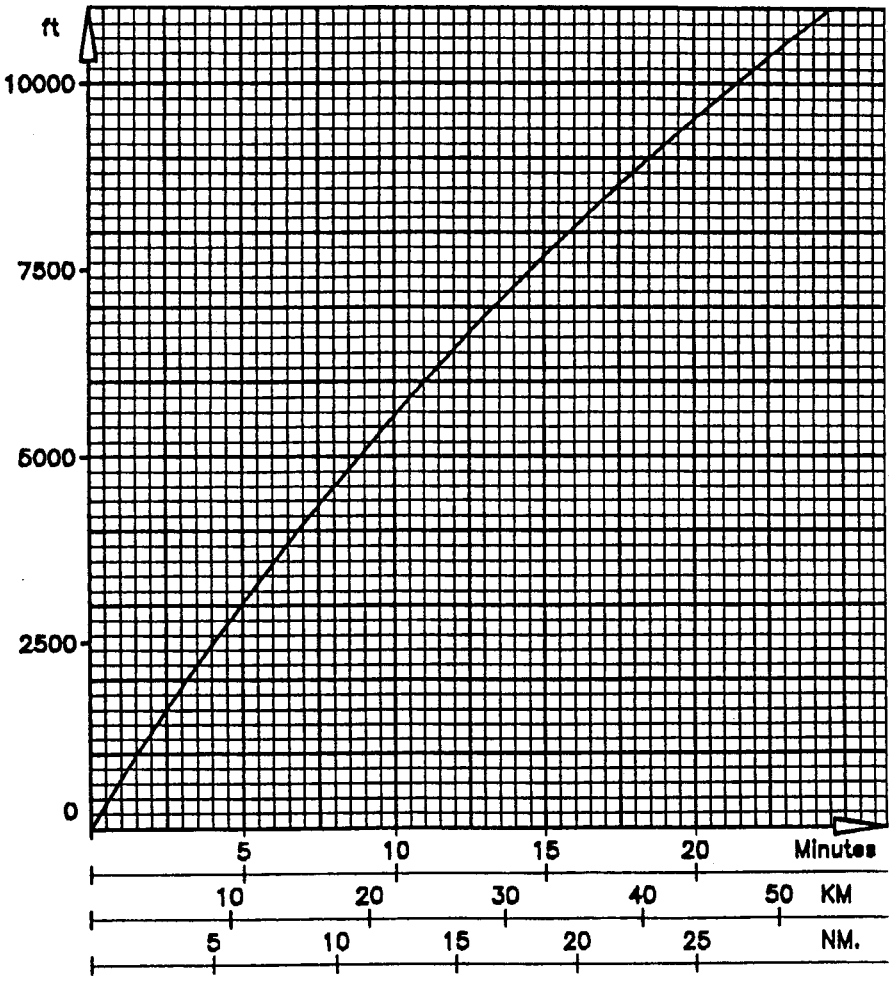
Take off from dry grass runway: add 8 %

CLIMB PERFORMANCE



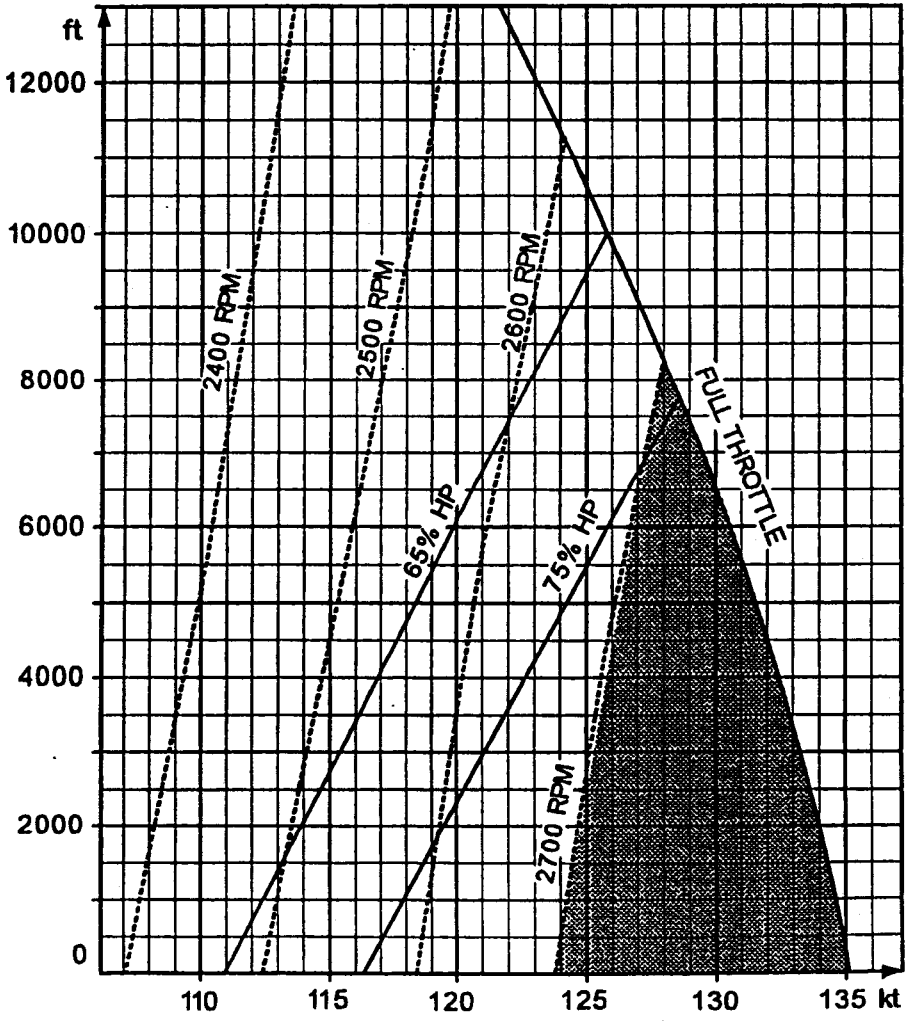
CLIMB TIME / CLIMB DISTANCE

Standard atmosphere
Flaps up
Full throttle
MTOW
Climb speed (IAS): 78 kt (145Km/h)
Consumption 30 l/h



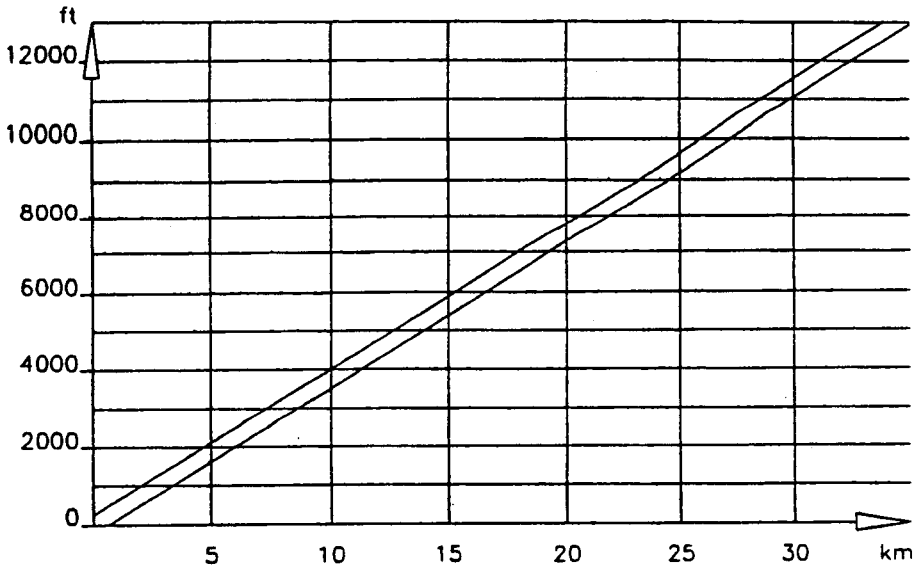
CRUISE PERFORMANCE

MTOW (1984 lb)
Flaps up
Standard atmosphere
No wind
Mixture at best power setting
64' Sensenich propeller



GLIDE PERFORMANCE

Airspeed..... (78 kt) 145 km/h
Propeller ..... windmilling
Flaps ..... up
Without wind



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Altitude and temperature do not have a perceptible influence.

LANDING PERFORMANCE

Dry, hard runway. Flaps 35°. power off							
Max weight kg (lb)	Head wind (kt)	Sea level + 15°C		2500 ft - 760 m + 10°C		5000 ft - 1525 m + 5°C	
		Run m (ft)	15 m (50 ft) clear	Run m (ft)	15 m (50 ft) clear	Run m (ft)	15 m (50 ft) clear
900 (1984)	0	233 (764)	440 (1444)	250 (820)	465 (1526)	265 (869)	490 (1608)
	10	165 (541)	365 (1198)	175 (574)	390 (1280)	185 (607)	413 (1365)
	20	120 (394)	295 (968)	130 (426)	310 (1017)	140 (459)	335 (1099)
800 (1764)	0	220 (722)	415 (1362)	235 (771)	440 (1444)	250 (820)	465 (1526)
	10	155 (509)	345 (1132)	165 (541)	370 (1214)	175 (574)	390 (1280)
	20	115 (377)	280 (919)	125 (410)	295 (968)	130 (426)	315 (1033)
700 (1543)	0	190 (623)	375 (1230)	205 (673)	400 (1312)	215 (705)	420 (1378)
	10	135 (443)	315 (1033)	145 (476)	335 (1099)	150 (492)	350 (1148)
	20	100 (328)	250 (820)	110 (361)	270 (886)	115 (377)	280 (919)

Landing on dry grass runway: increase distances by 20%.

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SECTION 6 WEIGHT AND BALANCE

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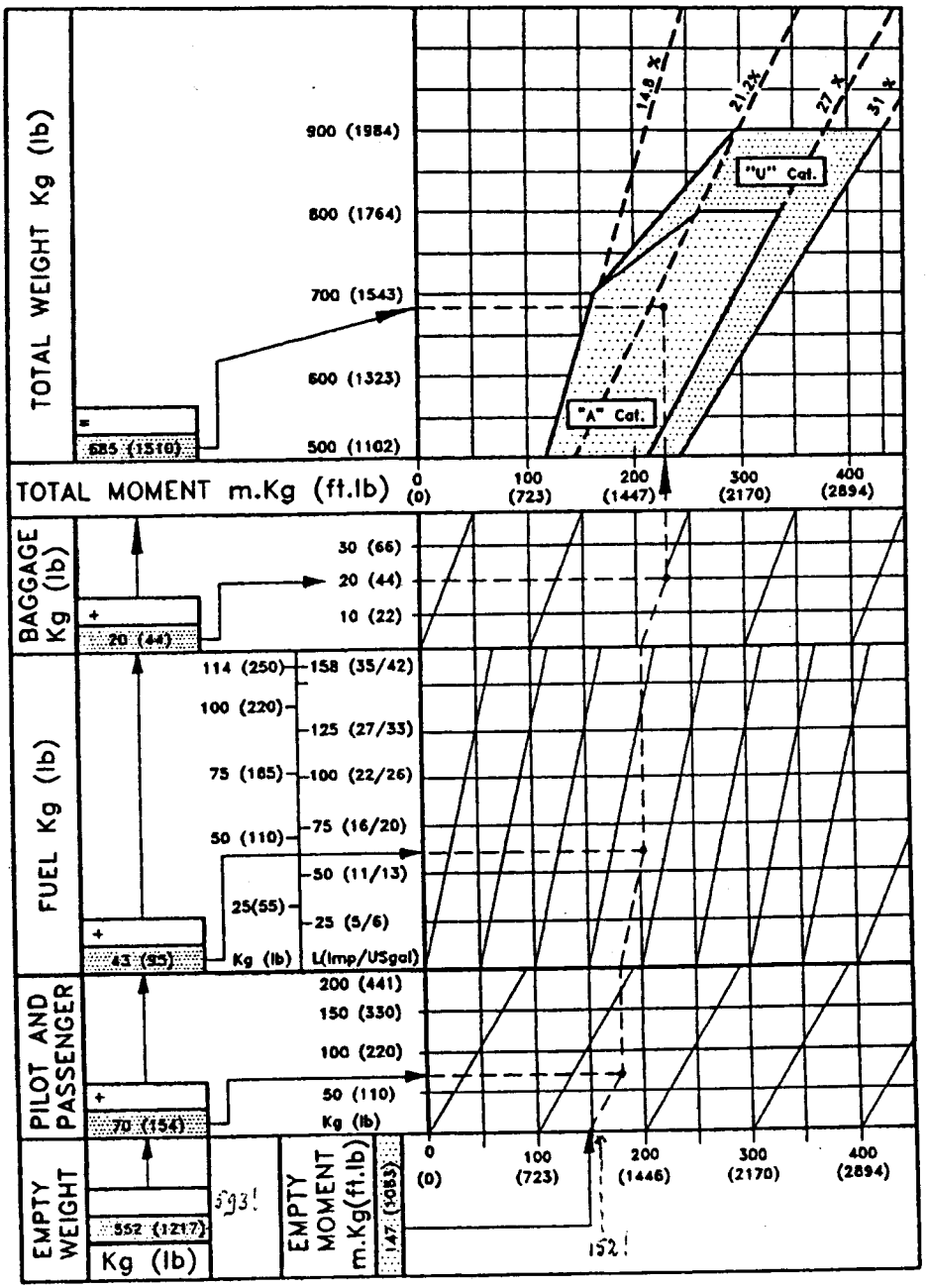
Use of the weight and balance diagram 6.03


TOTAL WEIGHT Kg (lb)
900 (1984)
800 (1764)
700 (1543)
600 (1323)
500 (1102)
685 (1510)
14.8 %
21.2%
27 %
31 %
"U" Cat.
"A" Cat.
TOTAL MOMENT m.Kg (ft.lb)
0 (0)
100 (723)
200 (1447)
300 (2170)
400 (2894)
BAGGAGE Kg (lb)
30 (66)
20 (44)
10 (22)
20 (44)
FUEL Kg (lb)
114 (250)
100 (220)
75 (165)
50 (110)
25(55)
Kg (lb)
158 (35/42)
125 (27/33)
100 (22/26)
75 (16/20)
50 (11/13)
25 (5/6)
L(Imp/USgal)
43 (95)
PILOT AND PASSENGER
200 (441)
150 (330)
100 (220)
50 (110)
Kg (lb)
70 (154)
EMPTY WEIGHT
552 (1217)
Kg (lb)
593!
EMPTY MOMENT m.Kg(ft.lb)
147 (1083)
0 (0)
100 (723)
200 (1446)
300 (2170)
400 (2894)
152!

USE OF WEIGHT AND BALANCE DIAGRAM

1) Calculate the total loaded aircraft weight:
   empty weight (from the weight and balance sheet)
   + pilot and passenger
   + baggage
   + standard fuel

Insure that total weight does not exceed 900 kg (1984 lb).

2) Place the empty aircraft moment (from the weight and balance sheet) on the upper scale of the opposite diagram, and follow the example indicated by the dashed line.
   The resulting point must be within the centre of gravity moment envelope (shaded area) for the load to be within limits.

EXAMPLE *

Empty aircraft moment	(825 ft.lb)	147 m.kg
Empty aircraft weight	(1217 lb)	552 kg
Pilot + passenger	(154 lb)	70 kg
Fuel 60 l (13 Imp/16 us gal)	(95 lb)	43 kg
Baggage	(44 lb)	20 kg
TOTAL WEIGHT	(1510 lb)	685 kg

CENTRE OF GRAVITY: within the envelope

1 liter AVGAS	= 0,72 kg	(1.6 lb)
1 imp gal AVGAS	= 3,27 kg	(7.2 lb)
1 us gal AVGAS	= 2,7 kg	(6 lb)

*** ATTENTION**

For your aircraft centre of gravity calculation, please do not use values of empty aircraft weight and empty aircraft moment indicated in the above example. Use values indicated on the last weight and balance sheet of your aircraft.

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SUPPLEMENT 1 NIGHT VFR

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SECTION 1 - DESCRIPTION

The R2160 aircraft can be used in Night VFR in non-icing conditions after implementation of "Modification majeure n° 34". This major modification provides for the installation of an instrument panel lighting system.

In order to be used in Night VFR, it is mandatory that the R2160 be fitted with the following equipment, in addition to the day VFR equipment:

Navigation flight

- Speed indicator;
- Sensitive altimeter, adjustable, pressure indication in hPa, 1 turn per 1000 ft
- Magnetic compass with deviation device;
- Vertical speed indicator;
- Gyro horizon;
- Turn and bank indicator;
- Second gyro horizon or gyro turn indicator;
- Directional gyro;
- V.O.R. or A.D.F. depending on the routing or an approved GPS class A, B or C;
- Handheld flashlight
- Spare fuses;
- Navigation lights;
- Anticollision light;
- Landing light;
- Instrument panel lighting

Communication

- a E/R V.H.F. 25 kHz;
- in H type zone, a H.F. transmitter-receiver;

Control

- a E.L.T. (R.D.B.A.)
- a transponder mode A.

SECTION 2 - LIMITATIONS

In order to be used in Night VFR, it is mandatory that the aircraft be fitted with the serviceable equipment required by the Decree dated July 24, 1991.

The limitations of section 2 remain unaffected by operating the aircraft in Night VFR, except the flight conditions placard, page 2.06, which must be replaced by the following placard:

THIS AIRCRAFT MUST BE USED IN ACROBATIC OR UTILITY CATEGORY IN ACCORDANCE WITH THE APPROVED FLIGHT MANUAL ON THIS AIRCRAFT, ALL PLACARDS CORRESPOND TO ACROBATIC UTILISATION. FOR UTILITY OPERATION, REFER TO THE APPROVED FLIGHT MANUAL. MANEUVERING SPEED 127 kt - 236 km/h VFR FLIGHT BY DAY OR BY NIGHT IN NON-ICING CONDITIONS NO SMOKING

SECTION 3 - EMERGENCY PROCEDURES

The following emergency procedures complete those of the Section 3.

Lighting 1 failure

Lighting 2 ..... on
Lighting 1 fuse ..... verify

If the failure persists, the lighting 2 and the torch can be used as an emergency lighting.

Light failure

Taxi lighting switch-type circuit breaker ..... verify

Battery failure

If, following a complete battery failure, the alternator deactivates involving loss of power supply, proceed as follows:

Battery, alternator and radio (if installed) circuit breakers ..... off
Battery switch ..... on
Alternator switch ..... on

Notice that circuits are powered again. Reset only those switches necessary to ensure flight safety.

Electrical power supply malfunction

Alternator failure is indicated when the amber "alternator failure" light on the warning panel is lit, and a progressive drop in voltage (shown on the voltmeter).

If amber light is lit

Switch off alternator, then back on. This operation resets the overvoltage relay which may have cut-out due to a transiant overvoltage.

If the failure continues

- Switch off the alternator
- Switch off all electrical equipment not essential for continuing the flight
- Land as soon as possible and have the electrical system inspected

NOTE

An alternator failure does not prevent the engine from operating normally

SECTION 4 - NORMAL PROCEDURES

These procedures complement those of the Section 4.

Preparation

- Study the weather reports in order to avoid flight in dangerous conditions (minima, icing...).
- Check that fuel and oil quantities comply with the flight plan necessities and the regulations.

Pre-flight check

Check the operation of the following equipment:

Anticollision light ..... verify
Navigation light ..... verify
Landing light ..... verify
Taxi light ..... verify
Cabin lighting ..... verify
Instrument panel lighting ..... verify
Day/night dimmer switch ..... verify
Emergency flashlight on board ..... verify

Lighting

- switch on lighting 2
- adjust with lighting 1 if necessary

Taxiing

Anticollision light ..... on
Navigation light ..... on
Gyro instruments ..... check by alternate turns
Artificial horizon ..... set index
Directional gyro ..... rotation correct
Turn and bank indicator ..... direction of movement correct

Before take off

Vacuum indication..... check
VHF ..... test
VOR or ADF or GPS..... test
Heating and defrosting ..... as needed
Landing light ..... on

Aligning

- Set directional gyro

Take off

- Maintain positive climb on rate of climb indicator.
- Switch off taxi and landing lights at the end of the runway.

Climb and cruise

Above 8000 feets, there is a risk of disturbance in the pilot's night vision..

Landing

Landing light ..... on
Taxi light ..... on

After engine shut down

Lights ..... off

SECTION 5 - PERFORMANCE

The performance described in Section 5 remain unaffected.

SUPPLEMENT 2 CENTURY II B AUTOPILOT

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Section 1 Description 7.10
Section 2 Limitations 7.14
Section 3 Emergency procedures 7.14
Section 4 Normal procedures 7.15
Section 5 Performance 7.16

SECTION 1 - DESCRIPTION

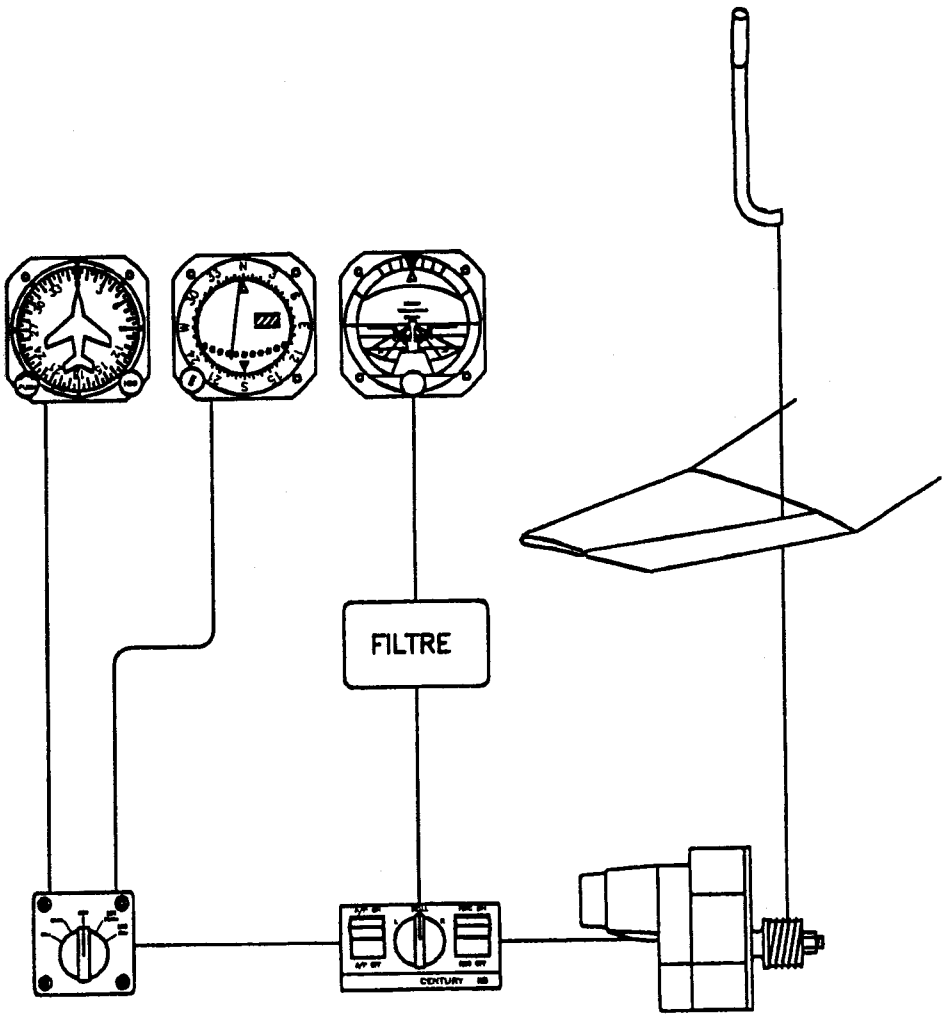
The CENTURY II B is an all electric, single axis (roll) autopilot system. It provides course intercept, tracking and coupler for VOR/ILS optional equipment.

CENTURY II B COMPONENTS DESCRIPTION

Command console

A/P ON - A/P OFF	Autopilot ON-OFF rocker switch. When only this switch is engaged (A/P ON), the autopilot is responsive to the roll command knob (ROLL) on the center of the console.
ROLL	Roll maneuvre knob up to approximately 30° of bank right or left. The centered position represents approximate wings level flight. When the heading mode is engaged (HDG ON), action on the roll knob (ROLL) is ineffective.
HDG ON - HDG OFF	Heading mode ON-OFF rocker switch. It provides turning the aircraft to a preselect heading on the Course Selector Directional Gyro or conducting course intercepts or tracking. When heading mode switch is engaged (HDG ON), roll knob input is replaced by the Course Selector Directional Gyro and the optional Coupler Mode Selector inputs. The Course Selector Directional Gyro and the optional Coupler Mode Selector should be set prior to engagement of the heading mode (HDG ON). (See Section on Coupler Mode Selector operations when optional Coupler Mode Selector is installed)

CENTURY II B AUTOPILOT INSTALLATION DIAGRAM



Course Selector Directional Gyro

Any heading may be selected, either before or after heading mode (HDG ON-HDG OFF) engagement on the command console and turns up to 160° may be programmed directly, either right or left.
If course selector indicator is rotated beyond 180° from the Directional Gyro card heading, the autopilot will turn in the shortest direction to reach the selected heading.
In normal operation the maximum bank in heading mode (HDG) is 20°.

Coupler Mode Selector

Direct the autpilot in both VOR and ILS navigation.

Nominal interception angles are 45° and an automatic 15° crosswind correction capability is provided.

"HDG" Mode — It is the basic function of the CENTURY II B autopilot as described in Command console section.

"OMNI" Mode — When in the "OMNI" Mode position the system is coupled to the VOR Indicator.
To select a desired course for interceptor tracking always set both the VOR Course Selector and the Directional Gyro Course Indicator to the desired course. All heading will then be controlled by the VOR signal. A full deflection on the VOR Indicator will produce a 45° interception angle.
With less than full deflection, the system will automatically direct a smooth, tangential intercept to arrive over the radial with crosswind correction established. The same dynamic intercept is accomplished whether 3 km (2 miles) or maximum reception distance from station.
Below approximately 3 km (2 miles), autopilot bank limitations will allow a slight overshoot of the selected radial.

"NAV" Mode

This mode serves the same functions as "OMNI" Mode and is initiated in the same manner.
The "NAV" Mode, however, incorporates an extended time delay which reduces reaction to short term VOR needle deflections.
The "NAV" Mode is recommended for enroute navigation or any time autopilot response to short term VOR needle deflections becomes excessive.
The "NAV" Mode should not be used for close in VOR approach work, as close in work requires the proportioned dynamic response provided by the "OMNI" Mode.

"LOC NORM" Mode

In the "LOC NORM" Mode, the system adjusts its sensitivity to accomodate the "Localizer" course width (5° instead of 20° for the VOR) and to provide smoother more optimum intercept and track maneuvers.
Intercept angles of 45° are automatic with tangential intercepts outside the marker and automatic crosswind correction.
As with the "OMNI" Mode, the Course Selector Directional Gyro must be set to correspond the desired magnetic course.

"LOC REV" Mode

The features of "LOC REV" Mode are identical to the "LOC NORM" Mode except that the aircraft will fly away from the Localizer Indicator Needle instead of toward it.
When using the "LOC REV" Mode the Course Selector Indicator must be set to the reciprocal of the front course heading.

SECTION 2 - LIMITATIONS

Section 2 limitations are not affected by the CENTURY II B autopilot installation .

Following autopilot specific limitations are to be added:

Minimum use height ..... 500 ft
Maximal use speed .....(140 kt) 260 km/h

IMPORTANT

Do not use the autopilot in case of Directional Gyro, vacuum pump or system failure.

SECTION 3 - EMERGENCY PROCEDURES

In case of autopilot malfunction:

1- Manoeuvrer the stick as required in order to override the autopilot

NOTE

The autopilot may be overriden by the pilot without damage to the system.

2- Cut off the main autopilot switch (AP/OFF)

3- Pull the autopilot breaker and do not try to switch it on again

In case of vacuum system failure:

1- Cut off the main autopilot switch (AP/OFF)

SECTION 4 - NORMAL PROCEDURES

Preflight autopilot check with engines running and gyros erected:

1 - Vacuum ..... green arc

2 - Autopilot switch ..... "A/P OFF"

3 - Heading mode switch ..... "HDG OFF"

4 - Coupler Mode Selector ..... "HDG"

5 - "ROLL" knob ..... centered

6 - Course Selector Directional Gyro ..... centered

7 - Autopilot switch ..... "A/P ON"

8 - Turn the "ROLL" knob ..... "L" then "R"
(note that the control stick responds in the proper direction)

9 - Heading mode switch ..... "HDG ON"

10 - Rotate the heading knob ..... "R" then "L"
(note the autopilot response. Without aerodynamic response, servo action is unlimited)

11 - Override the autopilot at the control stick ..... "R" then "L"
(Force required should be approximately 7 kg (15 lbs))

12 - Before take off autopilot switch ..... "A/P OFF"

In flight autopilot engagement procedure

Aircraft flight attitude ..... wings level

"ROLL" knob ..... centered

Heading mode switch ..... "HDG OFF"

Autopilot switch ..... "A/P ON"

Coupler Mode Selector ..... "HDG"

Heading knob ..... centered

Heading mode switch ..... "HDG ON"

Select desired lateral guidance mode

For more details concerning the use of available modes, refer to the CENTURY II B pilot's operating manual.

Final approach procedure

In final approach and at latest 500 ft height:

Autopilot switch ..... "A/P OFF"

SECTION 5 - PERFORMANCE

Section 5 performance are not affected by CENTURY II B autopilot installation.

SUPPLEMENT 3 AIRCRAFT WITHOUT WHEEL SPATS

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Section 1 Description 7.18
Section 2 Limitations 7.18
Section 3 Emergency procedures 7.18
Section 4 Normal procedures 7.18
Section 5 Performance 7.18
Section 6 Weight and balance 7.18

SECTION 1 - DESCRIPTION

The removal of the wheel spats is permitted.

SECTION 2 - LIMITATIONS

Unchanged.

SECTION 3 - EMERGENCY PROCEDURES

Unchanged

SECTION 4 - NORMAL PROCEDURES

Unchanged

SECTION 5 - PERFORMANCE

Take off performance
The 15 m (50 ft) clearance distance must be increased by 2,1%.

Climbing performance
The Vz must be decreased by 2%.

Cruising performance
Level flight speeds must be decreased by 6%.

SECTION 6 - WEIGHT AND BALANCE

The empty weight must be decreased of the wheel spats weight.

The variation of the empty centrage balance is insignificant.

SUPPLEMENT 4 GPS

TABLE OF CONTENTS

Section 1 General 7.20
Section 2 Limitations 7.20
Section 3 Emergency procedures 7.20
Section 4 Normal procedures 7.20
Section 5 Performance 7.20
Section 6 Weight and balance 7.20

SECTION 1 - GENERAL

The GPS mentioned hereunder are approved on R2160 series for day VFR in view of the ground or the sea.
The integrity of the position indicated by the GPS is not guaranteed. It is therefore the responsibility of the pilot to verify the accuracy of this position by all means of navigation available.
The GPS operating manual at latest edition must be on board the aircraft for use by the pilot.
The coupling of the GPS mentioned hereunder to a flight director or to an autopilot has not been demonstrated, except for the KLN90A equipment.

GPS APPROVED ON R2160	
GARMIN	150
KING	KLN90, KLN90A, KLX 135

in deze kist

SECTION 2 - LIMITATIONS

The following placard completes those of the page 2.06:

GPS LIMITED TO DAY VFR
IN VIEW OF THE GROUND OR THE SEA ONLY

SECTION 3 - EMERGENCY PROCEDURES: Unchanged

SECTION 4 - NORMAL PROCEDURES: Unchanged

SECTION 5 - PERFORMANCE: Unchanged

SECTION 6 - WEIGHT AND BALANCE: Unchanged

SUPPLEMENT 5 ENLARGED CANOPY

TABLE OF CONTENTS

Section 1 General 7.22
Section 2 Limitations 7.22
Section 3 Emergency procedures 7.22
Section 4 Normal procedures 7.22
Section 5 Performance 7.22
Section 6 Weight and balance 7.22

SECTION 1 - DESCRIPTION

In order to improve the habitability of the cockpit, the R2160 can be fitted with an enlarged canopy.

SECTION 2 - OPERATING LIMITS

Unchanged.

SECTION 3 - EMERGENCY PROCEDURES

Unchanged.

SECTION 4 - NORMAL PROCEDURES

Unchanged, except:

Spin

The spin recovery could equally be engaged by the following optional procedure:

throttle ..... idle
flaps ..... up
rudder..... fully opposed direction
elevator control ..... push forward until rotation stops
ailerons ..... neutral

SECTION 5 - PERFORMANCE

Unchanged.

SECTION 6 - WEIGHT AND BALANCE

See Weight data.

SUPPLEMENT 6	LOW VOLTAGE

Supplement 6
withdrawn
14-03-2000

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Section 2 Limitations 7.24
Section 3 Emergency procedures 7.24
Section 4 Normal procedures 7.24
Section 5 Performance 7.24
Section 6 Weight and balance 7.24

SECTION 1 - GENERAL

A steady warning light is installed which will illuminate if the alternator output reduces to a level where the battery supplies power to the bus-bar (see Airworthiness Notice n°88).

Before engine start
Check low volts warning..... ON

After engine start
Check low volts warning..... OFF

If warning illuminates during flight
Reduce electrical load
Battery duration approx..... 30 minutes
Land on the nearest suitable airfield

NOTE
Warning may illuminate with low engine rpm. Check it goes out when rpm increases.

SECTION 2 - LIMITATIONS: Unchanged

SECTION 3 - EMERGENCY PROCEDURES: Unchanged

SECTION 4 - NORMAL PROCEDURES: Unchanged

SECTION 5 - PERFORMANCE: Unchanged

SECTION 6 - WEIGHT AND BALANCE: Unchanged

Alternative spin recovery procedure supplement

TABLE OF CONTENTS

Section 1 General 2
Section 2 Limitations 2
Section 3 Emergency procedures 2
Section 4 Normal procedures 2
Section 5 Performance 2
Section 6 Weight and balance 2

SECTION 1 - DESCRIPTION

An alternative spin recovery procedure can be applied for CAA operated aircraft.

SECTION 2 - LIMITATIONS

The ~~Les~~ limitations of the Section 2 remain unchanged except the flight conditions placard pages 2.08 & 7.05 which must be replaced by the following one:

ACROBATIC CATEGORY APPROVED MANEUVERS

SPIN (flaps up)	54 KIAS	CHANDELLE	120 KIAS
POSITIVE LOOP	130 KIAS	HALF LOOP & ROLL OUT	135 KIAS
ROLL	108 KIAS	FLICK ROLL	86 KIAS
STALL TURN	120 KIAS	LAZY EIGHT	120 KIAS

SPIN RECOVERY PROCEDURE

THROTTLE CLOSED
FLAPS UP
RUDDER IN FULLY OPPOSITE SPIN DIRECTION
ELEVATOR CONTROL FORWARD TO NEUTRAL
AILERONS NEUTRAL

SECTION 3 - EMERGENCY PROCEDURES: Unchanged

SECTION 4 - NORMAL PROCEDURES

Unchanged, except:

Spinning

Following spin recovery procedure may also be applied:
Throttle..... closed
Flaps..... up
Rudder..... fully opposite spin direction
then elevator..... forward to neutral
Ailerons..... neutral

SECTION 5 - PERFORMANCE: Unchanged

SECTION 6 - WEIGHT AND BALANCE: Unchanged